

Design and Implementation of a Compiler with Machine-Learning-Guided Optimization Selection

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Abstract—This paper documents the complete design, implementation, and evaluation of MiniLang, a research compiler for a minimal statically typed imperative language. MiniLang was build from scratch in C++17 to support end-to-end study of compiler construction and machine-learning-guided optimization. The system implements a classical multi-phase pipeline: hand-written lexical analysis, recursive-descent parsing, two-pass semantic analysis, lowering to a flat High-level Intermediate Representation (HIR), configurable rule-based optimizations, textual LLVM IR emission, and an ML-ready optimization advisor with evaluation harness.

We describe every compilation phase in implementation-level detail, including token design, abstract syntax tree (AST) structure, semantic rules, HIR instruction set, three optimization passes, LLVM code generation strategy, and a feature-driven advisor that selects optimization pipeline using heuristics, JSON plans or external Python interface. Experimental results in benchmark programs show that optimization reduces the HIR instruction count by up to 63% ($22 \rightarrow 8$ instruction) and LLVM IR size by 41% ($34 \rightarrow 20$ lines) on constant-heavy workloads, while preserving correct runtime behavior (verified by process exit codes.) Minilang is intended as a reproducible research and teaching traditional compiler engineering and modern ML-for-compilers research

Index Terms—Compilers, intermediate representation, LLVM, static analysis, code optimization, machine learning, pass ordering

I. INTRODUCTION

A. Motivation

Optimizing compilers transform source programs through dozens of analysis and transformation passes. Deciding which passes to run, and in what order has been tuned manually for decades in systems such as GCC and LLVM[1],[2]. Recently, machine learning (ML) has been applied to predict profitable compiler decisions—including pass ordering, heuristics flags, and inlining thresholds—from program features[3]–[6]. However, experimenting with ML-guided optimization typically requires navigating industrial-strength compiler infrastructures whose complexity obscures the relationship between front-end structure, IR design, and optimization outcomes.

Minilang addresses this by providing a complete but minimal compiler where every phase is inspectable, independently executable, and instrumented for measurement. This project follows a staged research methodology:

- Language design - define a small typed language sufficient for control flow, functions, and arithmetic.

- Front-end construction- implement lexer, parser, and semantic analyzer with explicit error reporting.
- IR design - lower AST to flat HIR with explicit temporaries and structured control flow.
- Rule-based optimization - implement classic local optimizations at HIR level.
- Code generation - emit human-readable LLVM IR for execution via Clang.
- ML advisor integration - extract static features, select pass pipelines, and evaluate against baselines.

B. Research Questions

This work investigates:

- RQ1: Can a minimal imperative language support a full, pedagogically transparent compilation pipeline comparable to textbook architecture [1],[7]?
- RQ2: Is a flat HIR with explicit temporaries a suitable layer for both rule-based optimization and feature extraction?
- RQ3: Can static HIR metrics drive selective application of optimization passes with measurable IR reduction?

C. Contributions

- Formal specification of the MiniLang language (lexical, syntactic, and semantic rules).
- Complete architectural documentation of a C++17 compiler with six standalone driver tools.
- HIR specification with lowering algorithms for expressions, conditionals, and loops.
- Three HIR optimizations with fixed-point iteration and statistics.
- ML-ready advisor framework: 14-dimensional feature vector, pass, registry, heuristic/JSON/Python modes, and minilang_eval harness.
- Empirical evaluation demonstrating IR reduction and correctness on benchmark programs.

D. Paper Organization

Section II reviews related work. Section III defines MiniLang. Section IV presents system architecture. Sections V–X detail each compilation phase. Section XI covers the ML advisor. Section XII presents evaluation. Section XIII–XIV discuss limitations and conclude.

II. RELATED WORK

A. Compiler Construction Foundations

The classic compiler pipeline- lexical analysis, syntax analysis, semantic analysis, intermediate code generation, optimization and code generation- is established in Aho et al.[1] and Cooper and Torczon [2]. MiniLang adopts this structure verbatim, using a hand-written scanner and recursive-descent parser rather than generator tools(Flex/Bison) to maximize transparency for readers and students.

Appel [8] and Muchnick [9] emphasize the role of intermediate representation in decoupling front-end language features from back-end code generation. Our HIR follows the principle: it is lower than AST(flat instruction list) but higher than LLVM (no SSA no target- specific details).

B. LLVM as a Back-end

Lattner and Adve introduced LLVM as a modular compilation framework with SSA-based intermediate representation and aggressive optimization infrastructure [2]. Rather than linking against the LLVM C++ API initially, MiniLang emits textual LLVM IR (.ll files), which Clang consumes for native code generation. This approach mirrors the LLVM Kaleidoscope tutorial [10] and keeps the research artifact readable in published papers.

C. Machine Learning for Compiler Optimization

Ashouri et al. provide a comprehensive survey of ML applied to compiler autotuning, pass ordering, and heuristic selection [3]. CompilerGym exposes LLVM optimization as a reinforcement-learning environment [4]. Other work learns graph representations of programs for heuristic tuning [5],[6]. MiniLang differs from these systems in scope and interpretability: optimization decisions occur at HIR level with explicit, low-dimensional features (14 integers), making it suitable for classroom demonstration and lightweight ML prototyping without requiring GPU-scale training infrastructure. The advisor is architected as a pluggable policy (heuristic today; ONNX or Python model tomorrow), aligning with survey recommendations for separating feature extraction from inference [3].

D. Educational Compilers

Educational compilers traditionally prioritize clarity over performance. MiniLang extends the educational pattern by adding (1) a documented semantic analyzer with symbol tables (2) a serialized HIR dump format, and (3) quantitative evaluation tooling- bridging teaching and research.

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	Table column subhead	Subhead	Subhead
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^aSample of a Table footnote.



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ACKNOWLEDGMENT

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REFERENCES

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